



The wet planet?

No longer is that title reserved only for Earth. With the discovery of water on Mars, everyone with an interest in space is stoked. Here's what they're talking about.

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The fourth rock from the sun has always been known to us as a wasteland. Sure there have been a few movies and books that romanticise it and try and make it some fantasy-world filled with life. Total Recall actually tried to suggest that aliens had left the means to terraform Mars into an Earth-like planet, and many of us want that to be accomplished as well. However, everything we found out about Mars, courtesy all the missions by NASA and others, left us a tad disappointed. The hope of even finding microbial life on Mars was dwindling, and fast. Until now.

On September 28, NASA held a press conference to announce that liquid water does indeed flow on today's Mars. Scientists had earlier identified streaks on slopes of craters and hills as probably caused by flowing water, but the question was always "When?". As in, were these streaks made last week, a couple of centuries ago, or 3 million years ago? Were the streaks actively growing, or were they just remnants of a past history of water on Mars? These streaks were studied in detail, and termed recurring slope lineae (RSL) by Lujendra Ojha, a researcher from Georgia Tech, Atlanta, who was born in Nepal.

NASA confirmed that these RSL were active, and that in Mars' summer, there is



Lujendra Ojha of Nepal was instrumental in studying the RSL phenomenon, and played a huge part in NASA's discovery of flowing water on Mars

actually liquid water inside these gullies, which seeps down the slopes.

Pool party!

Sadly, there isn't enough liquid water at any one place to do that as a human, but we're all very interested to see whether there are any microscopic organisms who are skinny-dipping in this Martian water.

Of course, you should know that this water was discovered by analysing a type of salt that was found in the gullies and streaks – called perchlorates, some forms of which were detected to contain water in their crystals. Perchlorates are a type of hydrated salt (contains water), that not only captures water, but also modifies its

characteristics significantly. For example, given that the Martian atmosphere is so thin, and the pressure is so low, liquid water (which boils at about 100 degrees centigrade on Earth) would begin to boil at around 10 degrees on Mars. Pure water would still freeze at 0 degrees, making the effective temperature range for liquid water 0 to 10 degrees (or 10 times less than Earth's 0 to 100 degrees). Thankfully, the perchlorates step in here, and reduce the freezing temperature of the water they capture to minus 70 degrees centigrade. This effectively gives the briny water that seeps down the slopes on Mars a much better 80+ degree range to exist, and that's why we see those streaks.

Mars attacks!

So how likely are we to find life inside those streaks of seeping water? Although some people are ever the optimists and feel if there's liquid water, there is always a chance of life existing, sadly, the majority seem to think that those little streaks of seeping water will just be way too salty to ever support life (as we know it). Most bacterial life cannot survive in freezing temperatures with that amount of salt content. The closest comparison we have here on Earth to this type of water on Mars is at the Don Juan Pond in Antarctica. Widely accepted as the saltiest water body on Earth, Don Juan almost never freezes because it is so salty. It's estimated to have over 40% salinity level, which is significantly greater